

UTTARA UNIVERSITY

School of Science & Engineering

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



Syllabus for B. Sc. (Engineering) in Computer Science and Engineering (B.Sc. in CSE) Four Year Program

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UTTARA UNIVERSITY
School of Science & Engineering
Department of Computer Science & Engineering
Syllabus

for
Bachelor of Science (Engineering) in Computer Science & Engineering (B.Sc. in CSE)

Program Name: Bachelor of Science (Engineering) in Computer Science and Engineering [B.Sc. in CSE]

Name of the School: School of Science and Engineering

Name of the Department: Computer Science and Engineering

Program Objectives:

The objective of the Bachelor of Science in Computer Science and Engineering (B.Sc. in C.S.E) Program is to produce world class professionals in the field. The program covers a comprehensive curriculum following standard international guidelines. The program has been designed so as to achieve the perfect blend of theory and practice in order to provide a path for students to steadily increase their engineering knowledge and skills. It is expected that with this program the students will get adequate training in a wide variety of software system environments and in an equally wide variety of hardware platforms.

Students will gain knowledge in software development. Internet programming, Network engineering, Computer architecture, Microprocessor, VLSI (Very Large Scale Integration) Software and hardware engineering to equip themselves appropriately to face the challenges of the emerging computer industry in the 21st century. Besides, following the recent trends in advanced universities in the developed world, The CSE curriculum includes some selected interdisciplinary subjects. These subjects form a general body of knowledge along with the major area of concentration in order to prepare the students to adjust to the reality of contemporary world facing multidimensional problems and issues.

Course Category

Letter symbols for course categories are:

CSEC – Computer Science and Engineering
EENG – Electrical and Electronic Engineering
MATH – Mathematics
PHY – Physics
ACTG - Accounting
ECON - Economics

Semesters

There will be three semesters in an academic year, which are:

- **Spring-** January to April
- **Summer-** May to August
- **Fall-** September to December

Duration of the Program:

The duration for B.Sc. in Computer Science and Engineering program is four years divided into 12 semesters.

Credit-hour:

Three credits are assigned to a theory course if there are three hours lecture in a week; one and a half credits for three hours practical/sessional course work in a week.

Degree/ Graduation Requirement:

A student need to complete minimum of 153 credit hours. For any special case may get some course or/and semester waiver will be decided by the appropriate authority of Uttara University. Passing of all courses individually and maintaining a minimum CGPA of 2.0.

Degree to be awarded:

Bachelor of Science (Engineering) in Computer Science and Engineering

Entry Requirement

For Bachelor of Science (Engineering) in Computer Science and Engineering (or B.Sc. in CSE) program, the entry level is a Higher Secondary School Certificate (HSC) background, or a 4-year Diploma in Engineering or any other equivalent level of education. Entry level will be decided by the appropriate authorities of Uttara University for the students from foreign institutes.

Admission

Admission into Bachelor of Computer Science and Engineering (CSE) program is on a competitive basis. Minimum second division in both S.S.C. and H.S.C. with science background or under GPA system will have to have a minimum CGPA of 2.5 in separately or minimum CGPA 6 in total. The O- and A-level students must have an average grade of B. Candidates are selected for admission on the basis of past academic records, a written test and a viva-voice examination.

Grading System and Performance Evaluation

Students having minimum 75% class attendance will be eligible to take semester examinations. Letter grading will be made to assess students' performance. The grade will be assigned on the overall evaluation of a student's performance on the basis of semester final examination, midterm exam, Class test/ Quizzes, term papers, assignment and class attendance in aggregate. The teachers responsible for the course will determine Grades/GPA. The final result will be prepared by cumulating the grade point average (CGPA) over the courses. The UGC approved common grading system is adopted for assigning a letter grade and grade point. This is given in the following table:

Numerical	Letter Grade		Grade point	Comments
80% and above	A+	(A Plus)	4.0	Outstanding
75% to less than 80%	A	(A regular)	3.75	Excellent
70% to less than 75%	A-	(A Minus)	3.50	Very good
65% to less than 70%	B+	(B Plus)	3.25	Good
60% to less than 65%	B	(B regular)	3.00	Above average
55% to less than 60%	B-	(B Minus)	2.75	Average
50% to less than 55%	C+	(C+ Plus)	2.50	Below average
45% to less than 50%	C	(C regular)	2.25	Fair
40% to less than 45%	D		2.00	Pass

Less than 40%	F		0.00	Fail
	F * Failure I ** Incomplete W *** Withdrawal R ***** Repeat Y ***** Audit			

- * “F” means failure. Credits for courses with this grade do not apply towards graduation.
- ** “I” grade is given to students who have fulfilled the majority of the course requirements but have been unable to complete the rest. The student is not required to register for the course in the next semester.
- *** “W” means withdrawal. A student may decide to withdraw from a course by the deadline with the consent of the instructor and the Academic Advisor.
- ***** “R” means repeat. If a student obtains a grade lower than ‘B’ in a course, he/she will be allowed to repeat the course only once for the purpose of grade improvement by forgoing his/her earlier grade, but he/she will not be eligible to get a grade better than ‘B+’ in such a course. A student will be permitted to repeat for grade improvement purpose a maximum of four courses in B.Sc. Engineering program.

If a student obtains ‘B’ or a better grade in any course, he/she will not be allowed to repeat the course for the purpose of grade improvement.

Marking System in the Individual Courses

Continuos assesment	- 30%
Midterm Examination	- 30%
Final Examination	- 40%

Marks Distribution for Continuous Assessment

Thirty percent (30%) of marks shall be allotted for continuous assessment i.e., in class tests/quizzes and class participation. Basis for awarding marks for class participation will be as follows:

Attendance	Marks
90% and above	10
85% to less than 90%	9
80% to less than 85%	8
75% to less than 80%	7
70% to less than 75%	6
65% to less than 70%	5
Less than 65%	0

Total No. of class tests/quizzes shall be at least $n+1$, where n is the number of credits of the course. Evaluation of the performance in class tests/quizzes will be on the basis of the best n class tests/quizzes. The scheme of continuous assessment will be announced on the first day of classes. **To qualify for participation in the examinations, a student must have class attendance at least 65%.**

Workload

The minimum number of credit hours that a student must take per semester is 12 and the maximum is 18. This may be relaxed only with special permission of appropriate authorities of Uttara University.

Course Requirements

The Course Requirements for the B.Sc. (Engg.) in Computer Science & Engineering program:

Table-1

COURSES	NUMBER OF COURSES	TOTAL CREDIT HOURS
Compulsory Theoretical Courses	28	84
Compulsory Sessional Courses	19	28.5
Elective –I	Any 03 courses can be chosen from 08 courses	13.5
Elective – II	Any 02 courses can be chosen from 12 courses	6
Compulsory Interdisciplinary Courses	5	15
Project Work	2	6
Total Credit	59 (Courses)	153 (Credit hours)

The course requirements for Bachelor of Science (Engineering) in Computer Science & Engineering Program are as follows:

Compulsory Theoretical and Sessional Courses (Total 112.5 Credits)

Course Code and Name

COURSE CODE	COURSE NAME	CREDIT HOURS
CSEC 111	Computer System and Application	3
CSEC 112	Computer System and Application Sessional	1.5
CSEC 121	Programming Language and Application-I	3
CSEC 122	Programming Language and Application-I Sessional	1.5
CSEC 211	Programming Language and Application-II	3
CSEC 212	Programming Language and Application-II Sessional	1.5
CSEC 213	Basic Electronics	3
CSEC 214	Electrical and Electronic Technology Sessional	1.5
CSEC 215	Data Structure	3
CSEC 216	Data Structure Sessional	1.5
CSEC 217	Digital Logic Design	3
CSEC 218	Digital Logic Design Sessional	1.5
CSEC 225	Microprocessor, Interfacing and Low Level Language	3
CSEC 226	Microprocessor, Interfacing and Low Level Language Sessional	1.5
CSEC 227	Operating System Design	3
CSEC 228	Operating System Design Sessional	1.5
CSEC 229	Computer Architecture	3
CSEC 311	Design and Analysis of Algorithm	3
CSEC 312	Design and Analysis of Algorithm Sessional	1.5

CSEC 313	Object - Oriented Programming	3
CSEC 314	Object - Oriented Programming Sessional	1.5
CSEC 316	Systems Analysis and Design	3
CSEC 317	Systems Analysis and Design Sessional	1.5
CSEC 318	Computer Peripherals, Interfacing and Maintenance	3
CSEC 319	Computer Peripherals, Interfacing and Maintenance Sessional	1.5
CSEC 321	Database Management System	3
CSEC 322	Database Management System Sessional	1.5
CSEC 323	Data Communication	3
CSEC 324	Measurement and Instrumentation	3
CSEC 325	Compiler Construction	3
CSEC 326	Compiler Construction Sessional	1.5
CSEC 428	Computer Graphics	3
CSEC 429	Computer Graphics Sessional	1.5
CSEC 430	Software Engineering	3
CSEC 431	Software Engineering Sessional	1.5
CSEC 432	Computer Communication and Networks	3
CSEC 433	Computer Communication and Networks Sessional	1.5
MATH 110	Differential & Integral Calculus	3
MATH 123	Discrete Mathematics	3
MATH 129	Complex Variable & Laplace Transforms	3
MATH 221	Linear Algebra and Fourier Transformation	3
MATH 462	Numerical Analysis and Coordinate Geometry	3
STAT 121	Statistics and Probability	3
EENG 121	Electrical Circuits	3
EENG 122	Electrical Circuits Sessional	1.5
PHY 110	Modern Physics	3
PHY 111	Modern Physics Sessional	1.5

Elective-I (Any three from eight courses, 3 X 4.5 = 13.5 credits)

COURSE CODE	COURSE NAME	CREDIT HOURS
CSEC 422	Internet Programming & Sessional	4.5
CSEC 423	VLSI Design & Sessional	4.5
CSEC 424	Artificial Intelligence and Expert Systems & Sessional	4.5
CSEC 425	Simulation, Modeling and Performance Analysis & Sessional	4.5
CSEC 426	Pattern Recognition & Sessional	4.5
CSEC 427	Digital Pulse Techniques & Sessional	4.5
CSEC 466	Multimedia Theory & Sessional	4.5
CSEC 467	Computer Vision and Image processing & Sessional	4.5

Elective – II (Any two from twelve courses, 2 X 3 = 6 credits)

COURSE CODE	COURSE NAME	CREDIT HOURS
CSEC 411	Fault Tolerant Systems	3
CSEC 412	Management Information Systems	3
CSEC 413	Digital Control Systems	3

CSEC 414	Digital Signal Processing	3
CSEC 415	Operations Research	3
CSEC 416	Computer Security	3
CSEC 417	Computer System Design	3
CSEC 457	Graph Theory	3
CSEC 418	Machine Learning	3
CSEC 419	Theory of Computing	3
CSEC 459	Mathematical Analysis for Computer Science	3
CSEC 421	Human Computer Interaction in Bangladesh	3

Interdisciplinary Compulsory Courses (Total 15 Credits)

COURSE CODE	COURSE NAME	CREDIT HOURS
ENG 100	Communication Skill	3
BUS 110	Business Communication	3
ACTG 121	Accounting	3
ECON 229	Economics	3
MNGT 228	Management and Organization Behavior	3

Project (Total 6 Credits)

COURSE CODE	COURSE NAME	CREDIT HOURS
CSEC 399	Project-I	3
CSEC 499	Project-II	3

Semester wise course distribution:

Year-1, Semester-1					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC 111	Computer System and Application	3	3	0	
CSEC 112	Computer System and Application Sessional	1.5	0	1.5	
MATH 110	Differential & Integral Calculus	3	3	0	
ENG 100	Communication Skill	3	3	0	
STAT 121	Statistics and Probability	3	3	0	
Total	=	13.5	12	1.5	

Year-1, Semester-2					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC 121	Programming Language and Application-I	3	3	0	
CSEC 122	Programming Language and Application-I Sessional	1.5	0	1.5	
MATH 129	Complex Variable & Laplace Transforms	3	3	0	
MATH 123	Discrete Mathematics	3	3	0	
EENG 121	Electrical Circuits	3	3	0	
EENG 122	Electrical Circuits Sessional	1.5	0	1.5	
Total	=	15	12	3	

Year-1, Semester-3

COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC 211	Programming Language and Application-II	3	3	0	
CSEC 212	Programming Language and Application-II Sessional	1.5	0	1.5	
BUS 110	Business Communication	3	3	0	
ACTG 121	Accounting	3	3	0	
PHY 110	Modern Physics	3	3	0	
PHY 111	Modern Physics Sessional	1.5	0	1.5	
Total	=	15	12	3	

Year-2, Semester-1					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC 213	Basic Electronics	3	3	0	
CSEC 214	Electrical and Electronic Technology Sessional	1.5	0	1.5	
CSEC 215	Data Structure	3	3	0	
CSEC 216	Data Structure Sessional	1.5	0	1.5	
CSEC 217	Digital Logic Design	3	3	0	
CSEC 218	Digital Logic Design Sessional	1.5	0	1.5	
Total	=	13.5	9	4.5	

Year-2, Semester-2					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
ECON 229	Economics	3	3	0	
MATH 221	Linear Algebra and Fourier Transformation	3	3	0	
CSEC 229	Computer Architecture	3	3	0	
CSEC 225	Microprocessor, Interfacing and Low Level Language	3	3	0	
CSEC 226	Microprocessor, Interfacing and Low Level Language Sessional	1.5	0	1.5	
Total	=	13.5	12	1.5	

Year-2, Semester-3					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC 227	Operating System Design	3	3	0	
CSEC 228	Operating System Design Sessional	1.5	0	1.5	
CSEC 311	Design and Analysis of Algorithm	3	3	0	
CSEC 312	Design and Analysis of Algorithm Sessional	1.5	0	1.5	
CSEC 313	Object - Oriented Programming	3	3	0	
CSEC 314	Object - Oriented Programming Sessional	1.5	0	1.5	
Total	=	13.5	9	4.5	

Year-3, Semester-1					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC 316	Systems Analysis and Design	3	3	0	
CSEC 317	Systems Analysis and Design Sessional	1.5	0	1.5	
CSEC 321	Database Management System	3	3	0	
CSEC 322	Database Management System Sessional	1.5	0	1.5	

CSEC 323	Data Communication	3	3	0	
CSEC 324	Measurement and Instrumentation	3	3	0	
Total	=	15	12	3	

Year-3, Semester-2					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC 318	Computer Peripherals, Interfacing and Maintenance	3	3	0	
CSEC 319	Computer Peripherals, Interfacing and Maintenance Sessional	1.5	0	1.5	
CSEC 325	Compiler Construction	3	3	0	
CSEC 326	Compiler Construction Sessional	1.5	0	1.5	
CSEC 428	Computer Graphics	3	3	0	
CSEC 429	Computer Graphics Sessional	1.5	0	1.5	
Total	=	13.5	9	4.5	

Year-3, Semester-3					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
MATH 462	Numerical Analysis and Coordinate Geometry	3	3	0	
CSEC 430	Software Engineering	3	3	0	
CSEC 431	Software Engineering Sessional	1.5	0	1.5	
CSEC 432	Computer Communication and Networks	3	3	0	
CSEC 433	Computer Communication and Networks Sessional	1.5	0	1.5	
Total	=	12	9	3	

Year-4, Semester-1					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
Elective –I (Any three from the following eight courses):					
CSEC 422	Internet Programming & Sessional	4.5	4.5	0	
CSEC 423	VLSI Design & Sessional	4.5	4.5	0	
CSEC 424	Artificial Intelligence and Expert Systems & Sessional	4.5	4.5	0	
CSEC 425	Simulation, Modeling and Performance Analysis & Sessional	4.5	4.5	0	
CSEC 426	Pattern Recognition and Sessional	4.5	4.5	0	
CSEC 427	Digital Pulse Techniques & Sessional	4.5	4.5	0	
CSEC 466	Multimedia Theory & Sessional	4.5	4.5	0	
CSEC 467	Computer Vision and Image processing & Sessional	4.5	4.5	0	
Total	=	13.5	4.5	0	

Year-4, Semester-2					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
MNGT 228	Management and Organization Behavior	3	3	0	
Elective –II (Any two from the following twelve courses):					
CSEC 411	Fault Tolerant Systems	3	3	0	
CSEC 412	Management Information Systems	3	3	0	

CSEC 413	Digital Control Systems	3	3	0	
CSEC 414	Digital Signal Processing	3	3	0	
CSEC 415	Operations Research	3	3	0	
CSEC 416	Computer Security	3	3	0	
CSEC 417	Computer System Design	3	3	0	
CSEC 418	Machine Learning	3	3	0	
CSEC 419	Theory of Computing	3	3	0	
CSEC 421	Human Computer Interaction in Bangladesh	3	3	0	
CSEC 457	Graph Theory	3	3	0	
CSEC 459	Mathematical Analysis for Computer Science	3	3	0	
Project work					
CSEC 399	Project-I	3	3	0	
Total	=	12	12	0	

Year-4, Semester-3					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC 499	Project-II	3	3	0	
Total	=	3	3	0	

Description of Courses:

CSEC 111 : COMPUTER SYSTEM AND APPLICATION

The Objective of this course is to teach fundamentals of computing, classification of computer, use the computer, scope advantage of computer, future of the computer information technology, Basic concept of Binary number system, Software, Hardware, System software, Application software, Networking Software.

Basic programming concepts; Program development stages: flow charts; Programming constructs: data types, operators, expressions, statements, control statements, functions, and array.

This course is designed in such a way that one will get first-hand knowledge on Operating system, Application Software, Graphics, Presentation, Database Management System, Networking and Basic level of Programming.

CSEC 112: COMPUTER SYSTEM AND APPLICATION SESSIONAL

Laboratory works based on '*CSEC 111: Computer System and Application*'

CSEC 121: PROGRAMMING LANGUAGE AND APPLICATION –I

The course acquaints the students with role and uses of computer with respect to programming language and decision making. It introduces the students to the concept of computer programming using C.

The course includes data types, operators, expressions, control structures; Functions and program structure: parameter passing conventions, scope rules and storage classes, recursion; Header files; Preprocessor; Pointers and arrays; Strings; Multidimensional array; User defined data types: structures, unions, enumerations; Input and Output: standard input and output, formatted input and output, file access; Variable length argument list; Command line parameters; Error Handling; Graphics; Linking; Library functions.

The course also includes problem analysis, programming algorithms, flow chart construction and programming on diverse areas.

CSEC 122: PROGRAMMING LANGUAGE AND APPLICATION -I SESSIONAL

Sessional works based on '*CSEC 121: Programming Language and Application –I*'

CSEC 211: PROGRAMMING LANGUAGE AND APPLICATION –II

This course has been designed to provide an introduction of both structured and object-oriented programming techniques using the C++ Language; this will equip the students with solid foundation and understanding of the skills of C++/ visual C++ or relevant (except JAVA) language enabling them to undertake the programming tasks competently and independently.

CSEC 212: PROGRAMMING LANGUAGE AND APPLICATION –II SESSIONAL

Sessional works based on '*CSEC 211: Programming Language and Application –II*'

CSEC 217: DIGITAL LOGIC DESIGN

Number systems and codes. Digital logic: Boolean algebra. De-Morgan's law, logic gates and their truth tables, canonical forms, combinational logic circuits, minimization techniques. Arithmetic and data handling logic circuits, decoders and encoders, Multiplexers and demultiplexers. Combination circuit design, Diode logic gates, transistor switches, transistor gates, MOS gates, Logic families, TTL, ECL, IIL, and CMOS logic with operation details, Propagation delay, Product and noise immunity, Open collector and High impedance gates, Electronic circuits for flip-flops, race around problems.

counters: asynchronous counters, synchronous counters and their PLA design. Synchronous and more machines, State minimization and assignments. Fundamental mode design A/D, D/A converters with applications. S/H circuits, LED, LCD and optically coupled oscillators, Non linear applications of OP Amps, Analogue switches.

CSEC 218: DIGITAL LOGIC DESIGN SESSIONAL

Sessional works based on '*CSEC 217: Digital Logic Design*'

CSEC 313: OBJECT-ORIENTED PROGRAMMING

The objective of this course is to acquaint students with oops concepts evolution and benefits. based on the skills gathered on JAVA computer the students will learn the programming technique using structured data types arrays, and pointers Lists, classes, Objects, static and dynamic polymorphism, inheritance etc. Using the features of DDE and OLE the students will learn how to connect to database performing queries transaction procession and error handling. GUI design and file access is also covered in this module.

(Pre-requisite: '*CSE 123: Programming Language and Application-II*')

CSEC 314: OBJECT-ORIENTED PROGRAMMING SESSIONAL

Sessional works based on '*CSEC 313 Object-Oriented Programming*'

STAT 121 : STATISTICS AND PROBABILITY

Basic probability theory, random variables: distribution and density functions, Gaussian random variable, transformation of a random variable, joint distribution and density, conditional distributions; expectation and moments; random processes: correlation functions, process measurements, Gaussian and Poisson random processes, noise models, power spectral densities: correlation and power spectrum, cross spectral densities; filtering of random processes, parameter estimation: maximum likelihood estimation, maximum a posteriori estimation, mean square error.

PHY 110 : MODERN PHYSICS

This course has been designed to teach the students in learning techniques and acquiring the skills needed for Heat and thermodynamics. Laws of thermodynamics Optics. Combination of Lenses. Theories of light, Interference of Light, Diffraction of Light Waves and Oscillations. Architectural Acoustics, Properties of matter, Atomic Structure of Matter, Elasticity, Viscosity, Hydrodynamics, Modern Physics, Relativity, Quantum effect, Mechanics, Radioactivity, Electromagnetism, Magnetic Principles of matter.

PHY 111 : MODERN PHYSICS SESSIONAL

Sessional works based on '*PHY 111: Modern Physics*'

EENG 121: ELECTRICAL CIRCUITS

The objective of this course is to teach students fundamentals of Electrical Engineering and measuring units including principles of D.C. voltage, current, resistance and power. Laws of electrical circuits and methods of network analysis, Single phase AC circuit analysts, DC measuring apparatus, Laws of magnetic fields and methods of solving simple magnetic circuits etcetera.

Network theorem and application compiled Circuits, self and mutual inductance, transient analysis of simple circuits, Polyphase circuit analysis and power measurement. Single phase transformers, equivalent circuits, three phase transformers.

Functions of Generators: Motors, Transformer, UPS, Voltage Stabilizer etcetera.

EENG 122: ELECTRICAL CIRCUITS SESSIONAL

Sessional works based on '*EEE 121: Electrical Circuits*'

MATH 110: DIFFERENTIAL & INTEGRAL CALCULUS

Differential Calculus: Limit, Continuity and differentiability. Successive differentiation of various types of functions, Leibnitz's theorem, Rolle's theorem, Mean value theorems, Taylor's and Maclaurin's theorems in finite and infinite forms, Lagrange's form of remainders, Cauchy's form of remainders. Expansion of functions, Evaluation of indeterminate forms by L'Hopital rule, Partial differentiation, Euler's theorem, Tangent and normal Concavity of functions, Determination of maximum and minimum values of functions and points of inflection with Applications. Curvature, Asymptotes

Integral Calculus: Integration by the method of substitution, Standard integrals, Integration by successive reduction, Definite integrals its properties and use in summing series. Walli's formula, Improper integrals, Beta function and Gamma function, Area under a plane curve and area of a region enclosed by two curves in Cartesian and polar co-ordinates.

MATH 123: DISCRETE MATHEMATICS

The course includes elements of Discrete Mathematics, Set theory, elementary number theory, Graph theory, Generating functions, Algebraic Structures, Serigraph, Permutation groups, Lattices, Finite fields and coding theory, Mathematical Logic, propositional calculus and predicate calculus. Theorem proof, relations, functions.

MATH 129 : COMPLEX VARIABLE & LAP LACE TRANSFORMS

Complex variable: Complex number System, general functions of a complex variable, Limits and continuity of a function of complex variable and related theorems, complex differentiation and the Cauchy-Riemann equations Mapping by elementary, Line integral of a complex function, Cauchy's integral theorem, Cauchy's integral formula, Liouville's theorem, Taylor's and Laurent's theorem, Singular points, Cauchy's Residue theorem, Evaluation of residues, Contour integration, Conformal mapping. Laplace Transforms Definition Laplace transforms of some elementary functions. Sufficient conditions for existence of Laplace transforms of derivatives, the unit step function, periodic function, some special theorems on Laplace transforms. Partial fraction Solutions of differential equations by Laplace transforms, Evaluation of improper integrals.

MATH 221: LINEAR ALGEBRA & FOURIER TRANSFORMATION

Matrices: Definition of matrix, Different types of matrices. Algebra of matrices adjoint and inverse of a matrix Rank and elementary transformations of matrices Normal and canonical forms Solution of linear equation matrix polynomials. Eigenvalues and eigenvectors.

Vectors: Scalars and vectors Equality of vectors, Addition and subtraction of vectors, Multiplication of vectors by scalars scalar and vector product of two vectors and their geometrical interpretation. Triple products and multiple products Linear dependence and independence of vectors. Differentiation and integration of vectors together with elementary applications, Definition of line, surface and volume integrals. Gradient, Divergence and curl of point functions, Various formula, Gauss's theorem, Stoke's theorem, Green's theorem.

Fourier Analysis: Real and complex form: Finite transform, Fourier integral, Fourier transforms and their uses in solving boundary value problems.

MATH 462 : NUMERICAL ANALYSIS AND COORDINATE GEOMETRY

Solution of linear and one-liner equations Linear Programming approximations of function, curve fitting interpolation extrapolation, integration differentiation, Solution of ordinary differential equations. This course offers laboratory practice and program implementation of various techniques of numerical methods. Coordinate geometry, Change of axes pair of straight-lines and System of circles.

CSEC 213: BASIC ELECTRONICS

The objectives of this course is to teach students introduction to Basic Electronics, Semiconductors. Junction diode characteristics, Bipolar transistor characteristics, Small Signal low frequency h-Parameter model Amplifiers the Darlington opera. Introduction to oscillators different ion amplifiers operational amplifiers linear applications of OP Amps gain input and crooned output impedance. Off-set null adjustments frequency response and noise Introduction to JFET, MOSFET, PMOS, NMOS and CMOS, Biasing and applications in switching circuits SCR, TRIAC, DIAC, UJT, Characteristics and applications. Introduction to rectifiers active filters regulated power supply, Basic idea about IC fabrication techniques.

(Pre-requisite: "EENG 121: Electrical Circuits")

CSEC 214: ELECTRICAL AND ELECTRONIC TECHNOLOGY SESSIONAL

Sessional works based on '*CSEC 213: BASIC ELECTRONICS*'

CSEC 215: DATA STRUCTURE

Concepts and examples, elementary data objects, elementary data structures, arrays, lists, stacks, queues graphs, trees, Memory management, sorting and searching hash techniques.

(Pre-requisite: '*CSEC 121: Programming Language and Application-I*')

CSEC 216: DATA STRUCTURE SESSIONAL

Sessional works based on '*CSEC 215: Data Structure*'

CSEC 225: MICROPROCESSOR, INTERFACING AND LOW LEVEL LANGUAGE

Introduction to different types of microprocessors.

Microprocessor architecture instruction set, interfacing I/O operation, Interrupt Structure, DMA, Microprocessor interface ICS, Review of interfaces ICS Bit slice microprocessor, Advanced Microprocessor in the control of Systems Use of simulators, Cross Compilers and development systems plus Low Level Programming.

(Pre-Requisite: '*CSEC 217: Digital Logic Design*')

CSEC 226: MICROPROCESSOR, INTERFACING AND LOW LEVEL LANGUAGE SESSIONAL

Sessional works based on '*CSEC 225: Microprocessor, Interfacing and Low Level Language*'

CSEC 229: COMPUTER ARCHITECTURE

A functional description of computer hardware, hardware concept and current technology. An investigation of Computer System design concepts including requirements, specifications, implementation and modification, Memories, buffers CPU characteristics, performance factors, overlay, parallel and pipeline system, virtual memory organization; IO Systems, Interrupt mechanism channels control units, input devices including telecommunication equipment.

(Pre-requisite: '*CSEC 217: Digital Logic Design*')

CSEC 227 : OPERATING SYSTEMS DESIGN

Principles of operating systems: design objectives, sequential processes, concurrent processes, concurrency, functional mutual exclusion, processor cooperation and deadlocks, processor management, Control and scheduling of large information processing systems. Resource allocation, dispatching, processor access methods, job control languages, Memory management, memory addressing, paging and store multiplexing. Multiprocessing and time sharing, batch processing, Scheduling algorithms, file systems, protection and security, design and implementation methodology, Performance evaluation and case studies, Windows NT: covers in detail both the server and work-station aspect of windows NT including features like administration and NT services. Multiuser Operation System Unix : covers UNIX from two different aspects as an operation system and a programming language using shell scripts It brings out the power and capability of UNIX in managing programs disks and other input/output devices.

CSEC 228 : OPERATING SYSTEMS DESIGN SESSIONAL

Sessional works based on '*CSEC 227: Operating Systems Design*'

CSEC 311: DESIGN AND ANALYSIS OF ALGORITHM

Techniques for analysis of algorithms, Methods for the design of efficient algorithms, divide and conquer, greedy method, dynamic programming, back tracking branch and bound, basic search and traversal techniques graph algorithms, algebraic simplification and transformations, lower bound theory NP , Hard and NP-complete problems.

(Pre-requisite: '*CSEC: 215 Data Structure*')

CSEC 312: DESIGN AND ANALYSIS OF ALGORITHM SESSIONAL

Sessional works based on '*CSEC 311: Design and Analysis of Algorithm*'

CSEC 316: SYSTEMS ANALYSIS AND DESIGN

Analysis of information gathering and editing, feasibility study operation, technical and economic feasibility, system design, tools and techniques, activity selections, Networks models, linear programming, project time estimation and minimization, cost benefit analysis, input output design, file design, database design, system implementation, project team selection, project maintenance, data mining techniques, telecommunications ethics, computer crime, security and control, documentation and report generation.

CSEC 317: SYSTEMS ANALYSIS AND DESIGN SESSIONAL

Sessional works based on '*CSEC 316 Systems Analysis and Design*'

CSEC 318 : COMPUTER PERIPHERALS, INTERFACING AND MAINTENANCE

The objective of this course is to familiarize the students with how to detect practical problems in computer devices. It will impart the practical knowledge of how to diagnose and solve the problems of input devices, processors, cache, RAM, ROM, Motherboard, different types of cards, storage media, display unit, printer and other computer-related equipments such as scanner OMR, bar code reader, UPS, stabilizer etcetera. Interface components and their characteristics, Optical displays and sensors, High power interface devices, transducers, stepper motors and peripheral devices.

(Pre-requisite: '*CSEC 229: Computer Architecture*' and '*CSEC 225: Microprocessor, Interfacing and low level Language*')

CSEC 319 : COMPUTER PERIPHERALS, INTERFACING AND MAINTENANCE SESSIONAL

Sessional works based on '*CSEC 318: Computer Peripherals, Interfacing and Maintenance*'

CSEC 321 : DATABASE MANAGEMENT SYSTEMS

The objective of this course is to acquaint the students with the concept of RDBMS. This module will enable the students to acquire skills to use a leading Relational Database Management System for distributed on-line transaction processing applications through the usage of Oracle, SQL, Database Libraries Visual Form/ Menu Generators etcetera.

CSEC 322 : DATABASE MANAGEMENT SYSTEMS SESSIONAL

Sessional works based on '*CSEC 321 : Database Management Systems*'

CSEC 323 : DATA COMMUNICATION

Fourier transforms Modulation techniques – AM, FM, PM, OOK, FSL, FSK, QPSK, QAM, Pulse modulation-PCM, PPM, PAM, Delta modulation. Compounding, Equalizers Echo cancellation Intersymbol, interference TDM, FDM, Error due to noise Concept of Channel coding and capacity Voice Digitization, Speech redundancies, DPCM Layered concept of computer network architecture.

CSEC 324 : MEASUREMENT AND INSTRUMENTATION

Ohmmeter, Voltmeter, Multi-meter, Multiplexes, Compurgator, Phase, period, frequency measurement, digital voltage measurement, instrumentation Characteristic, measurement of noise, recorders and display device, spectrum analyzer measurement and control of temperature, presser, displacement, density.

(Pre-requisite: '*CSEC 213: Basic Electronics*')

CSEC 325: COMPILER CONSTRUCTION

Introduction to compilers programming languages data elements and structures Operators, assignment Statements, parameter transmission, storage management, Finite automata and lexical analysis, syntactic specification of programming languages Basic parsing techniques, Automatic construction of efficient parsers syntax directed translation, symbols, error detection and recovery code optimization Compiler projects.

CSEC 326: COMPILER CONSTRUCTION SESSIONAL

Sessional works based on '*CSEC 325: Compiler Construction*'

CSEC 428: COMPUTER GRAPHICS

Introduction to Graphical data processing, Fundamentals of interactive graphics programming Architecture of display devices and connectivity to computer Implementation of graphics concepts of two-dimensional and three-dimensional viewing clipping and transformation hidden line algorithms, Raster graphics concepts Architecture, algorithms and other image synthesis methods Design of interactive graphic conversations.

(Pre-requisite: '*CSEC 121: Programming language and Application-I*')

CSEC 429 : COMPUTER GRAPHICS SESSIONAL

Sessional works based on '*CSEC 428: Computer Graphics*'

CSEC 430 : SOFTWARE ENGINEERING

Concepts of software engineering: requirements definition, modularity, structured design data specifications, functional Specification, verification, documentation software maintenance, software support tools, Software Project organization, quality assurance, management and organization quality assurance, management and communication skills. The course introduces software engineering and quality assurance covering the waterfall model including the various phased of analysis, design coding implementation, and maintenance management of software.

(Pre-requisite: '*CSEC 311: Design and Analysis of Algorithm*')

CSEC 431 : SOFTWARE ENGINEERING SESSIONAL

Software development using real life project (Based on '*CSEC 430: Software Engineering*')

CSEC 432 : COMPUTER COMMUNICATION AND NETWORKS

The objective of this course is to develop the students skill in computer networks (LAN) Types of network topologies network components including cabling layered architecture and operations TCP IP as the worldwide standard for communication across different platforms critical elements of TCP/IP including FTP & Telnet basic understanding of modem Repeaters, Bridges, Routers, Gateways and Telecommunications, Topological design and queuing models for network and distributed computing systems concept of managing network performance, Simple Network Management Protocol (SNMP) network data security.

(Pre-requisite: 'CSEC 323: Data Communication')

CSEC 433 : COMPUTER COMMUNICATION AND NETWORKS SESSIONAL

Sessional works based on 'CSEC 432: Computer Communication and Networks'

PROJECTS

CSEC 399: PROJECT-I

A project will be split into two parts Project-I and Project-II. In project-I, to study the problem area, formulate the research problem, identify critical areas, conduct Literature survey, study and analyze the problem for requirement definitions and design goals. The project may be hardware and software or both hardware and Software based system. One mini report will be submitted at the end of Project-I. The student will then continue with this project in Project-II for further development and implementation.

CSEC 499: PROJECT-II

Design, test, implement and study the performance of the solution for the problem in Project-I. Use of languages visual tools packages etcetera will be properly documented. A detailed project report should be submitted at the end of Project-II. The student will defend this project report with a presentation be a panel of project examiners.

ELECTIVE -I

CSEC 422: INTERNET PROGRAMMING & SESSIONAL

This course will acquaint the students with the most powerful and popular compiler used for the Internet, Java, and will cover advanced programming concepts such as programming with pages multiplatform programming and net-centric programming web page Developments and HTML, DHTML, XML, FLASH & Dreamweaver.

CSEC 423: VLSI DESIGN & SESSIONAL

Design and analysis techniques for VLSI circuits, Design of reliable VLSI circuits, Noise consideration, Design and operation of large fan out and fan in circuits , Clocking methodologies, techniques for data path and data control design, simulation techniques. This course will focus on the theoretical aspect of VLSI design only.

CSEC 424: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEM & SESSIONAL

Survey of concepts in artificial intelligence, knowledge representation, search and control techniques Machines and features of the LISP and PROLOG languages problem representation, search inference

and learning in intelligent systems for general problems solving. Game playing, expert consultation, concept formation and natural language processing recognition understanding and translation, Expert systems case studies.

(Pre-requisite: '*MATH 123: Discrete Mathematics*' and '*CSEC 311: Design and Analysis of Algorithm*')

CSEC 425: SIMULATION, MODELING AND PERFORMANCE ANALYSIS & SESSIONAL

Simulation methods model building, random number generator statistical analysis of results, validation and verification techniques Digital simulation of continuous systems Simulation and analytical method for analysis of computer systems and practical problems in business and practice, Introduction to the simulation packages, probability distribution and expectations stochastic processes, discrete time Markov chain and continuous time, Markova chain Birth death process in queuing suing models M/M/I, M/M/C, M/G/I, M/D/I, and G/M/I solutions of network of queues-closed queuing models and approximate models, Applications of queuing models in computer science and Engineering.

(Pre-requisite: '*STAT 121: Statistics and Probability*')

CSEC 426 : PATTERN RECOGNITION & SESSIONAL

Basic concepts Statistical pattern recognition, Discriminate functions Training Learning Syntactic pattern Recognition, Grammar-based approaches Recognition by passing Graphical approaches, learning via grammatical inference, Neural pattern recognition, Neural nets Pattern associations training by back propagation CAM approaches.

CSEC 427: DIGITAL PULSE TECHNIQUES & SESSIONAL

Diode logic gates, transistor switches, transistor gates, MOs gates Logic Families TTL, ECL, IIL and CMOS logic with operation details propagation delay Product and noise immunity open collector and high impedance gates, Electronic circuits for flip-flops, counters and register, memory systems, PLA's A/D and D/A converters with applications S/H circuits, LED, LCD and optically coupled oscillators, Non-linear applications of OP Amps Analogue switches.

(Pre-requisite; '*CSEC 213: Basic Electronics*')

CSEC 466: MULTIMEDIA THEORY & SESSIONAL

Overview of Multimedia Systems, Multimedia Storage, Data compression techniques for audio and video, Synchronization, Multimedia networking, and protocols, QOS principles, Video streams on ATM, Mobile – Multimedia communications, Operating System support for multimedia, Hypermedia system, Standard for Multimedia database and Multimedia Applications.

CSEC 467 : COMPUTER VISION AND PROCESSING & SESSIONAL

Introduction to Image Processing and Computer Vision: Representation of image, a basic image processing system, Relationship to human visual system, computer vision applications, Signal processing and pattern recognition.

Discrete Fourier Transform: One dimensional and pattern recognition digital images, two dimensional infinite Impulse Response Filters.

Two Dimensional System: Two dimensional signals frequency response of 2-D LSI systems, z transform, FIR filters, software implementation of 2-D FIR filters.

Techniques of image acquisition: CCD cameras, Scanners, Images Devices, Frame Grabbers, Gray levels, Coding of color pictures, binary patterns. Use of stereo cameras and Structured lighting for recognition of 3-D objects, use of range data for determination of depth, motion estimation, development of rule based system for image segmentation.

Image enhancement: Histograms equalization and modification, edge enhancement and detection, Region Growing Techniques for Image segmentation Textures, Line Tracing, Hough Transform for detection of lines and careless, recognition of 2D objects, Image Restoration and data compression: PSFs for different forms of blurs, estimating extent of blur and procedure for restoration, restoration through local operators, Document image processing, Data Compression, coding and decoding of digital image.

ELECTIVE-II

CSEC 411 : FAULT TOLERANT SYSTEMS

Introduction to fault Tolerant Systems and Architecture. Fault detection and location in combinational and sequential circuits: Digital simulation as a diagnostic tool. Automatic test pattern generator, Fault modeling, automatic test equipment, Faults in memory test pattern and reliability. Performance monitoring, self-checking circuits, Burst error correction and triple modular redundancy, Maintenance processors.

CSEC 412 : MANAGEMENT INFORMATION SYSTEMS

It will provide an integration framework of combining the technical aspects of MIS to the problems of managing complex organizations. The course highlights the issues involved and problem solving approaches thereof which are important for managing MIS projects and organizations. In addition, the initial stages of the system development process including feasibility study, requirement analysis, functional specification etcetera will also be covered.

CSEC 413 : DIGITAL CONTROL SYSTEMS

Introduction to control engineering, review of Laplace transform, transfer functions and systems response, controller design, stability of digital control systems, Signal conversion and processing, A-transform, state variable techniques, digital simulation, Syquest plots and Bode diagrams, design of digital control systems, Microprocessor control and experimentation.

CSEC 414 : DIGITAL SIGNAL PROCESSING

Applications and implementation of digital signal processing algorithms in areas such as multimedia and high-speed networks. Topics include discrete time signals and systems; discrete time Fourier transforms and Z-transforms, discrete Fourier transforms and fast implementations, digital filter design and implementations, and MultiMate signal processing Introduction to 2-dimensional signal processing, linear prediction, adaptive filtering, and applications in speech and image processing. The course will make use of MATLAB in Laboratory sessions.

CSEC 415 : OPERATIONS RESEARCH

The course is a survey of operations research techniques that has found application in the business area. Stress is placed upon model building, understanding and solution techniques including applications of

linear programming, transportation, assignment algorithms as well as sensitivity analysis of the variables, Other topics include game theory, queuing theory, simulation and sequencing models.

CSEC 416 : COMPUTER SECURITY

Computer security concepts, protecting computing resources against external threats, access control methods, theory and techniques of cryptography and encryption algorithms, digital signatures and standards, hash functions, network security, computer software and hardware used in cryptography, internet security.

CSEC 417 : COMPUTER SYSTEM DESIGN

Parallel computing architecture, mesh and hypercube architecture, parallel processing algorithms, computer arithmetic, CPU structure and functions, real time systems, Superscalar Processors, control unit operation, multiprocessing systems, Neuro-computing architectures.

CSEC 457 : GRAPH THEORY

An introduction to the theory of graphs, fundamental concepts and basic definitions, trees, spanning trees in graphs, distance in graphs, Eulerian graph, digraphs, matching and factors, cuts and connectivity, K-connected graphs, Manager's theorem, network flow problems, graph coloring: vertex coloring and edge coloring, line graphs, Hamiltonian cycles, planar graphs, perfect graphs.

CSEC 418 : MACHINE LEARNING

Introduction to Machine Learning: Supervised, Unsupervised and Reinforcement Learning: Unsupervised Learning Algorithms: Attribute Based and Relational Supervised Learning Algorithms; Neural Network based Learning Algorithms Genetic Algorithm and Genetic Programming; Reinforcement Learning Algorithms; Computational Learning Theory.

CSEC 419 : THEORY OF COMPUTING

Finite Automata: Deterministic finite automata, Nondeterministic finite automata, Equivalence and conversion of deterministic and nondeterministic finite automata, Pushdown automata Context Free Languages: Context tree grammars, Turing Machines: Basic machines, Configuration, Computing with Turing machines, Combining Turing machines.

CSEC 459: MATHEMATICAL ANALYSIS FOR COMPUTER SCIENCE

Recurrent Problems, Manipulation of Sums, Number Theory, Special Numbers, Generating Functions, Random Variables, Stochastic Process, Markov Chains (Discrete Parameter, Birth-Death Process), Queuing Models (Birth-Death Model, Markovian Model), Open and closed Queuing Network, Application of Queuing Models.

(Pre-requisite: 'STAT 121 : *Statistics and Probability*')

CSEC 421 : HUMAN COMPUTER INTERACTION IN BANGLADESH

Use of computer in Bangladesh. Scope of computer in Bangladesh. Feasibility to replace existing system. Communication system management, Prototyping, Simulation of modern concept, Product marketing customer behaviour, Implementation of automated system, Considerable factors to establish a new technology, Motivation, Entrepreneurship, Research about user interface.

INTERDISCIPLINARY COURSES

ENG 100 : COMMUNICATION SKILL

The purpose of this course is to develop the students' oral & written? Communication skills including communicative expressions for day to day actives. Both for personal and professional requirements, Grammar items will mainly emphasize the use of articles, number, tense, modal verbs, pronouns, question formation. Transformation of sentence. Simple passive voice construction conditionals will also be covered.

BUS 110 : BUSINESS COMMUNICATION

This course has been designed to help the students in learning the techniques and acquiring the skills needed to communicate effectively in business management. The course deals with various types of communications in the business environment. Various techniques of effective communication through business letters, memos, reports and other media form an integral part of the course. This will enhance the ability of the students in realistic business communication.

ACTG 121 : ACCOUNTING

The objective of this course is to deal with fundamentals of book-keeping and accounting. The course covers the basic accounting concepts, double entry system of bookkeeping, recording of transaction. Preparation of trial balance, treatment of adjustment entries, preparation of financial statements, accounting for assets and liabilities, accounting system and accounting practices in different types of organizations. Use of a spreadsheet program as a tool in undertaking various accounting tasks will be part of this course.

ECON 229 : ECONOMICS

The course highlights basic concepts of economics.

Microeconomics: It develops the students' skill in the systematic analysis of the economic aspects of decisions and acquaints them with some relevant analytical methods and concepts. The course covers concepts related to analysis of market as seen by producers and consumers, relates them to production patterns, distribution of income and allocation of resources in the context of a market economy.

Macroeconomics: It includes concepts, tools and techniques of macroeconomics, Emphasis is also given on national income and other related concepts like aggregate demand, supply, consumption and savings investment, demand for and supply of money.

MNGT 228 : MANAGEMENT AND ORGANIZATION BEHAVIOR

Management: This topic covers the functions of management, including planning, organizing, staffing, directing, motivating and controlling, problems in achieving organizational goals in different managerial environments. Each management principle and function is discussed in detail. **OB:** The objective of this topic is to deal with individual and group behavior models in the context of different organizations and social systems. It deals with the basic concepts of motivation, perception, learning and analysis of human behavior, individual differences and job satisfaction, attitude change, group process, team work, role theory, power and authority along with the analysis of small group behavior, group dynamics, leadership, decision, development of organizations and the influence of groups.

Uttara University
Dept. of Computer Science and Engineering
Semester-wise Course Distribution
B.Sc. in CSE (Evening)

Year-1, Semester-1					
COURSE CODE	COURSE TITLE	CREDIT	CREDIT HOURS		PREREQUISITE
			T	S	
CSEC111	Computer System & Application	3			
CSEC112	Computer System & Application Sessional	1.5			
CSEC121	Programming Language & Application I	3			
CSEC122	Programming Language & Application I Sessional	1.5			
ENG 100	Communication Skill	3			
EENG 121	Electrical Circuits	3			
Total		15			

Year-1, Semester-2					
CSEC211	Programming Language & Application II	3			
CSEC212	Programming Language & Application II Sessional	1.5			
MATH 123	Discrete Mathematics	3			
MATH 110	Differential & Integral Calculus	3			
CSEC 213	Basic Electronics	3			
CSEC 214	Electrical & Electronics Technology Sessional	1.5			
Total	=	15			

Year-1, Semester-3					
CSEC 215	Data Structure	3	3	0	
CSEC 216	Data Structure Sessional	1.5	0	1.5	
CSEC 229	Computer Architecture	3	3	0	
CSEC 217	Digital Logic Design	3	3	0	
CSEC 218	Digital Logic Design Sessional	1.5	3	0	
MATH 129	Complex Variable & Laplace Transform	3	0	1.5	
Total	=	15			
Year-2, Semester-1					

CSEC 227	Operating System Design	3			
CSEC 228	Operating System Design Sessional	1.5			
CSEC 323	Data Communication	3			
CSEC 225	Microprocessor Interfacing & Low Level Language	3			
CSEC 226	Microprocessor Interfacing & Low Level Language Sessional	1.5			
MATH 221	Linear Algebra & Fourier Transform	3			
Total	=	15			

Year-2, Semester-2					
CSEC 316	System Analysis & Design	3			
CSEC 317	System Analysis & Design Sessional	1.5			
CSEC 311	Design & Analysis of Algorithm	3			
CSEC 312	Design & Analysis of Algorithm Sessional	1.5			
CSEC 313	Object Oriented Programming	3			
CSEC 314	Object Oriented Programming Sessional	1.5			
STAT 121	Statistics and Probability	3			
Total		16.5			

Year-2, Semester-3					
CSEC 321	Database Management System	3			
CSEC 322	Database Management System Sessional	1.5			
MATH 462	Numerical Analysis and Coordinate Geometry	3			
CSEC 318	Computer Peripheral, Interfacing & Maintenance	3			
CSEC 319	Computer Peripheral, Interfacing & Maintenance Sessional	1.5			
CSEC 325	Compiler Construction	3			
CSEC 326	Compiler Construction Sessional	1.5			
Total		16.5			

Year-3, Semester-1					
CSEC428	Computer Graphics	3			
CSEC429	Computer Graphics Sessional	1.5			
CSEC430	Software Engineering	3			
CSEC431	Software Engineering Sessional	1.5			
CSEC 399	Project- I (Phase I)	3			
Elective-I : Any 01 Courses from the following 3 Courses (4.5 * 1 = Total 4.5 Credits)					
CSEC 422	Internet Programming & Sessional	4.5			
CSEC 423	VLSI Design & Sessional	4.5			
CSEC 426	Pattern Recognition & Sessional	4.5			
Total		16.5			

Year-3, Semester-2					
CSEC432	Computer Communication & Networks	3			
CSEC433	Computer Communication & Networks Sessional	1.5			
CSEC 399	Project- I (Phase II) [Note: It will not be registered in the ERP systems]				
Elective-I : Any 02 Courses from the following 5 Courses (4.5 * 2 = Total 9 Credits)					
CSEC 424	Artificial Intelligence and Expert Systems & Sessional	4.5			
CSEC 425	Simulation, Modeling and Performance Analysis & Sessional	4.5			
CSEC 427	Digital Pulse Techniques & Sessional	4.5			
CSEC 466	Multimedia Theory & Sessional	4.5			
CSEC 467	Computer Vision and Image processing & Sessional	4.5			
Total		13.5			

Year-3, Semester-3					
CSEC 499	Project-II	3			
Elective-II : Any 02 Courses from the following Courses (3 * 2 = Total 6 Credits)					
CSEC 411	Fault Tolerant Systems	3			
CSEC 412	Management Information Systems	3			
CSEC 413	Digital Control Systems	3			
CSEC 414	Digital Signal Processing	3			
CSEC 415	Operations Research	3			
CSEC 416	Computer Security	3			
CSEC 457	Graph Theory	3			
CSEC 418	Machine Learning	3			
CSEC 419	Theory of Computing	3			
CSEC 459	Mathematical Analysis for Computer Science	3			
Total		9			

Year-4, Semester-1

Any Student want to complete all the 153 credits courses then she/he has to continue another semester (132 +21=153) (10th Semester) (8 Courses)

PHY 110	Modern Physics	3			
PHY 111	Modern Physics Sessional	1.5			
CSEC 324	Measurement & Instrumentation	3			
EENG 122	Electrical Circuits Sessional	1.5			
BUS 110	Business Communication	3			
ACTG 121	Accounting	3			
ECON 229	Economics	3			
MNGT228	Management and Organizational Behavior	3			
Total		21			